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Trade Credit and the Stability of Supply Chains

Contents

1 Big picture

- **Question.** Does trade credit help stabilize production networks when a supplier becomes less reliable because of an operating shock?
- **Core idea.** An operating shock changes the value of a supplier relationship. Customers may switch away from a disrupted supplier, so the affected supplier can preserve the relationship by transferring surplus through more generous payment terms.
- **Network mechanism.** The response does not stop at the shocked firm. Its own suppliers also have a stake in the survival of the downstream relationship and may extend more trade credit to the affected firm so it can pass liquidity to customers.
- **Empirical shock.** The main shock is a natural disaster affecting the county of a firm's headquarters. Natural disasters proxy for temporary operating difficulties and are plausibly unrelated to the firm's own trade credit policy.
- **Main finding.** Firms affected by disasters simultaneously receive more trade credit from suppliers and extend more trade credit to customers.
- **Key magnitudes.** In the baseline firm-quarter panel, affected firms' accounts payable relative to COGS rises by about 8 percentage points, while accounts receivable relative to sales rises by about 2 percentage points.
- **Demand and supply evidence.** In hand-collected customer-supplier data, customers of affected firms receive more trade credit, and affected firms receive more trade credit from their own suppliers.
- **Mechanism 1.** Trade credit increases more when customers have lower switching costs, for example when the affected firm's products are easier to substitute or when customers are large.
- **Mechanism 2.** Trade credit increases less when customers expect lower disruption, for example when the affected firm has many locations, high inventories, long relationships, or superstar status.
- **Mechanism 3.** Trade credit flows increase more when the affected firm or its suppliers have more to lose from a relationship breakdown.
- **Financial constraints.** When the affected firm and its suppliers are financially constrained, trade credit cannot increase enough. Customers sever relationships and affected firms' sales and profitability fall.
- **Takeaway.** Trade credit acts like glue in supply chains: it reallocates surplus and liquidity after operating shocks, preserving valuable customer-supplier links.

2 Data and measurement

2.1 Production-network data

- The main sample combines Compustat financial data with FactSet Revere customer-supplier links.
- The unit of observation is a firm-quarter.
- The sample contains 107,106 firm-quarter observations in the summary table.
- The paper also constructs a hand-collected SEC sample from 10-K disclosures under FASB 105, which reports major-customer receivables and sales.
- The SEC sample gives supplier-customer-year observations, allowing the authors to identify trade credit flows between specific firms.

Interpretation: The FactSet/Compustat sample is broad and useful for firm-level dynamics; the SEC sample is narrower but much sharper about who gives trade credit to whom.

2.2 Natural disasters as operating shocks

- The shock variable equals one if a firm's headquarters county is hit by a natural disaster in a given quarter.
- Disasters come from SHELDUS data and include events with estimated damages above one billion 2012 dollars.
- These shocks include blizzards, earthquakes, floods, and hurricanes.
- Disasters are localized and affect an average of roughly 47 counties, with a maximum of 156 counties in a quarter.

Interpretation: The key identifying argument is that natural disasters impair operations and reliability without being chosen by the firm or its customers.

2.3 Trade credit variables

For affected firms, the two main firm-level measures are:

$$\Delta AP_{i,t,t+4} = \frac{AP_{i,t+4}}{COGS_{i,t+4}} - \frac{AP_{i,t}}{COGS_{i,t}}, \tag{1}$$

$$\Delta AR_{i,t,t+4} = \frac{AR_{i,t+4}}{Sales_{i,t+4}} - \frac{AR_{i,t}}{Sales_{i,t}}. \tag{2}$$

- Accounts payable measure trade credit received from suppliers.
- Accounts receivable measure trade credit extended to customers.
- The paper also studies net receivables, cash, assets, write-downs, COGS, sales, investment, ROA, customer entry/exit, and supplier choice.

Interpretation: Scaling payables by COGS and receivables by sales aligns trade credit with the relevant operating flow.

2.4 Mechanism variables

- **Switching costs:** product-market fluidity, industry HHI, and customer size.
- **Expected disruption:** number of states in which a firm operates and high inventory indicator.
- **Reputation:** relationship length and superstar status.
- **Supply-chain stakes:** relative customer size, major-customer indicators, supplier dependence on affected firms.
- **Financial constraints:** market capitalization below median and lack of bond rating.
- **External operating risk:** textual supply-chain risk from earnings calls.

Interpretation: These variables are chosen to map directly into the paper's three conceptual implications: customers' switching incentives, suppliers' incentives to preserve the chain, and constraints on liquidity provision.

3 Models and empirical specifications

3.1 Conceptual framework

The paper considers a simple supply chain. Firm 2 supplies Firm 1, and Firm 1 supplies Firm 0. Firm 1 is hit by an operating shock before it delivers inputs to Firm 0. Firm 0 can stay with Firm 1 or switch to another supplier at switching cost K.

- If Firm 0 stays, it may suffer disruption because the input is late or lower quality.
- If Firm 0 switches, Firm 1 and Firm 2 may lose sales and the supply chain may collapse.
- Firm 1 can offer trade credit to transfer surplus to Firm 0 and reduce the probability of switching.
- Firm 2 can extend trade credit to Firm 1 to make the downstream transfer feasible.

Interpretation: The framework treats trade credit as an incentive and surplus-transfer tool, not only as short-term financing.

3.2 Implication 1: affected firms extend trade credit to vulnerable customers

Affected firms extend more trade credit when their customers have low switching costs or when expected disruption is high:

$$\text{Trade credit from affected firm to customer} \uparrow \quad \text{if} \quad K \downarrow \quad \text{or expected disruption} \uparrow .$$

Interpretation: Customers who can easily leave require a larger transfer to remain in the relationship.

3.3 Implication 2: suppliers of affected firms accommodate liquidity

Suppliers of affected firms extend more trade credit when the affected firm's customers are likely to switch or when the supplier has more to lose from the chain's breakdown:

$$\text{Trade credit from upstream supplier to affected firm} \uparrow .$$

Interpretation: Upstream suppliers internalize the value of the whole production chain because losing the affected firm's customers ultimately hurts their own sales.

3.4 Implication 3: financial constraints weaken the trade-credit response

If affected firms or their suppliers are financially constrained, they cannot provide enough trade credit. Then:

$$\text{Trade credit response} \downarrow, \qquad \text{relationship termination} \uparrow, \qquad \text{sales and profits} \downarrow .$$

Interpretation: Trade credit can stabilize supply chains only if liquidity is available somewhere upstream.

3.5 Baseline difference-in-differences specification

The main regression is:

$$\Delta Y_{i,t,t+4} = \beta_0 + \beta_1 Affected_{it} + \beta_2 X_{i,t} + \eta_i + \eta_t + \eta_{j(i),y(t)} + \eta_{g(i),y(t)} + \varepsilon_{i,t}. \tag{3}$$

- $\Delta Y_{i,t,t+4}$ is the four-quarter change in a firm policy or performance variable.
- $Affected_{it}$ equals one if firm i is in a county hit by a natural disaster in quarter t.
- $X_{i,t}$ includes firm size, leverage, age, and profitability.
- Fixed effects include firm, year-quarter, industry-year, and state-year effects.

Interpretation: The coefficient β_1 captures the post-disaster change for affected firms relative to comparable firms, net of time, location, industry, and firm heterogeneity.

3.6 Customer-supplier trade-credit regressions

For suppliers of affected firms, the paper estimates:

$$\Delta y^{Received}_{i,s,y,y+1} = \beta_0 + \beta_1 Affected_{iy} + \eta_{s,y} + \epsilon_{i,s,y}. \tag{4}$$

For customers of affected firms, the paper estimates:

$$\Delta y^{Extended}_{i,c,y,y+1} = \beta_0 + \beta_1 Affected_{iy} + \eta_{c,y} + \epsilon_{i,c,y}. \tag{5}$$

Interpretation: These specifications compare trade credit provided to the same supplier or customer by affected versus unaffected firms, helping separate supply from demand.

3.7 Mechanism and constraint specifications

The generic interaction design is:

$$\Delta Y_{i,t,t+4} = \beta_0 + \beta_1 Affected_{it} + \beta_2 Z_{i,t} + \beta_3 (Affected_{it} \times Z_{i,t}) + FE + \epsilon_{i,t}. \tag{6}$$

- Z can be fluidity, HHI, number of states, high inventory, relationship length, superstar status, relative customer size, major-customer status, or supplier financial constraints.
- β_3 is the mechanism coefficient.

Interpretation: A positive β_3 for customer dependence means that affected firms with more to lose do more to preserve relationships. A negative β_3 for supplier financial constraints means that constrained suppliers block the stabilizing trade-credit response.

4 Identification and empirical design

4.1 Why natural disasters are useful

- Disasters are plausibly exogenous to firms' trade-credit choices.
- They are localized and temporary, allowing affected firms to be compared with unaffected firms in the same period.
- The empirical design includes rich fixed effects to absorb firm, time, state-year, and industry-year variation.

Interpretation: The empirical setting approximates a shock to operating reliability rather than a shock to financial demand.

4.2 Why this is not simply a liquidity-shock story

- A liquidity shock should make a firm demand more trade credit from suppliers and provide less trade credit to customers.
- The paper finds the opposite pattern for customer credit: affected firms both receive and provide more trade credit.
- This simultaneous increase is consistent with supply-chain stabilization, not a pure financing-shortfall story.

Interpretation: The sign pattern distinguishes operating shocks from liquidity shocks.

4.3 Robustness and pre-trends

- Table 4 shows no comparable pre-disaster trend in trade credit.
- Table 3 shows the effects survive alternative fixed effects and controls for affected customers or suppliers.
- SEC customer-supplier data confirm that actual bilateral trade-credit flows move in the predicted direction.

Interpretation: The evidence is strongest because the same mechanism appears in both broad firm-quarter data and bilateral customer-supplier disclosures.

4.4 Main limits

- Headquarters-county disasters may not perfectly capture all operational disruptions.
- Accounts receivable and payable are aggregate balance-sheet items in the main sample.
- The SEC bilateral sample is smaller and depends on disclosure of major customers.
- The study cannot observe every informal renegotiation of payment terms.

Interpretation: The paper addresses these limits by combining datasets and by testing multiple cross-sectional predictions.

5 Tables and figures

The paper contains no standalone figures. The notes therefore directly crop and group the original tables from the paper.

5.1 Table 1: Summary statistics

Read:

- The main Compustat/FactSet Revere panel has 107,106 firm-quarter observations.
- Mean change in payables is 0.015, and mean change in receivables is 0.014.
- The disaster dummy has mean 0.022, showing that disaster exposure is rare but not negligible.
- The SEC customer-supplier sample has 2,279 observations.

Economic message: The table establishes the scale of the two samples and the baseline magnitude of trade-credit changes.

Table 1
Summary statistics.

	N [1]	Mean [2]	SD [3]	P25 [4]	P50 [5]	P75 [6]
Panel A: Compustat-Factset Revere						
Change in payables	107,106	0.015	0.955	−0.070	0.003	0.076
Change in receivables	107,106	0.014	0.356	−0.052	0.001	0.058
Change in net receivables	107,106	−0.001	0.473	−0.064	0.000	0.065
Change in cash	107,106	−0.003	0.075	−0.031	0.000	0.030
Change in assets	107,106	0.061	0.224	−0.045	0.040	0.144
Change in write down	107,106	0.000	0.027	0.000	0.000	0.000
Change in log sales	107,106	0.061	0.231	−0.053	0.053	0.167
Change in investment	107,106	−0.047	0.793	−0.266	−0.002	0.213
Change in COGS	107,106	0.052	0.244	−0.061	0.051	0.169
Change in ROA	107,106	−0.001	0.027	−0.008	0.000	0.007
Disaster dummy	107,106	0.022	0.148	0.000	0.000	0.000
Size	107,106	6.989	1.907	5.621	7.243	8.659
Leverage	107,106	0.240	0.220	0.041	0.209	0.362
Age	107,106	2.645	0.996	2.079	2.773	3.367
Profit	107,106	−0.007	0.064	−0.008	0.009	0.020
Fluidity	84,996	6.854	3.305	4.298	6.214	8.724
HHI	107,106	0.144	0.119	0.055	0.101	0.196
Number of states	107,106	11.652	14.219	1	5	17
High inventory	107,106	0.495	0.500	0	0	1
Relationship length	107,106	3	1.985	1.667	2.546	4
Superstar	107,106	0.027	0.161	0	0	0
Relative size	107,106	1.531	2.822	−0.809	1.297	3.745
Major customer	107,106	0.254	0.435	0	0	1
Percentage of constrained suppliers (mkt cap)	106,656	0.096	0.195	0	0	0.115
Percentage of constrained suppliers (bond rating)	106,656	0.443	0.319	0.200	0.429	0.667
Change in number of customers	94,623	0.028	0.316	−0.098	0	0.160
Change in number of new suppliers	82,911	−0.034	0.493	−0.208	0	0.140
Panel B: SEC sample supplier-customer level						
Change in trade	2279	−0.005	0.073	−0.039	−0.003	0.031

Original Table 1: Summary statistics.

5.2 Tables 2–3: Main firm-level effects

Read:

- Table 2 shows affected firms' payables rise by 0.080 and receivables rise by 0.020.
- Net receivables, cash, assets, sales, COGS, and investment do not show comparable large changes in the baseline table.
- Table 3 confirms receivables and payables increase under richer fixed effects and controls.
- The effects are stronger for payables than receivables, consistent with liquidity flowing upstream-to-downstream through the affected firm.

Economic message: Affected firms do not simply borrow from suppliers because of a liquidity squeeze. They receive more trade credit and pass some of it on to customers.

Table 2
The impact of natural disasters on firms' policies.

	Change in payables	Change in receivables	Change in net receivables	Change in write down	Change in cash	Change in assets	Change in log sales	Change in COGS	Change in investment
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
Affected	0.080***	0.020***	0.001	0.002***	−0.002	0.002	0.005	−0.006	0.011
	(0.024)	(0.007)	(0.012)	(0.001)	(0.001)	(0.003)	(0.004)	(0.005)	(0.021)
Size	−0.003	−0.015***	0.005	−0.000	−0.005***	−0.151***	−0.066***	−0.056***	−0.035***
	(0.015)	(0.005)	(0.007)	(0.000)	(0.001)	(0.005)	(0.004)	(0.004)	(0.009)
Leverage	−0.091	−0.022	0.039	−0.001	0.024***	−0.132***	−0.037***	−0.066***	0.050
	(0.070)	(0.021)	(0.032)	(0.002)	(0.005)	(0.015)	(0.012)	(0.013)	(0.031)
Age	0.010	−0.006	−0.026***	0.000	0.010***	−0.010	−0.022***	−0.023***	0.051***
	(0.021)	(0.007)	(0.010)	(0.000)	(0.002)	(0.007)	(0.006)	(0.006)	(0.014)
Profit	−0.078	0.340***	−0.449***	−0.079***	−0.069***	0.165***	−0.295***	0.311***	0.296***
	(0.166)	(0.047)	(0.083)	(0.010)	(0.009)	(0.032)	(0.029)	(0.028)	(0.075)
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year-quarter FE	YES	YES	YES	YES	YES	YES	YES	YES	YES
State x year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES
Industry x year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES
Observations	93,702	93,702	93,702	93,702	93,702	93,702	93,702	93,702	93,702
R-squared	0.136	0.132	0.135	0.135	0.162	0.402	0.345	0.294	0.138

This table reports estimates of the effects of natural disasters on firms' policies. The unit of observation in each regression is a firm-quarter. The dependent variables are changes in the firm characteristics indicated on top of each column, defined as the difference between the value of the characteristic in quarter $t+4$ and t. For example, we define the change in accounts payable as $AP_{t+4}/COGS_{t+4} - AP_t/COGS_t$. The main independent variable is *Affected*, which is an indicator variable that equals one if a

Table 3
The impact of natural disasters on firms' receivables and payables.

	Change in receivables [1]	Change in receivables [2]	Change in receivables [3]	Change in receivables [4]	Change in receivables [5]	Change in payables [6]	Change in payables [7]	Change in payables [8]	Change in payables [9]	Change in payables [10]
Affected	0.023***	0.022***	0.019**	0.019**	0.019**	0.084***	0.084***	0.080***	0.079***	0.080***
	(0.008)	(0.008)	(0.007)	(0.008)	(0.008)	(0.023)	(0.024)	(0.024)	(0.025)	(0.025)
Size	−0.010**	−0.010**	−0.015***	−0.020***	−0.020***	−0.006	−0.006	−0.003	−0.001	−0.001
	(0.005)	(0.005)	(0.005)	(0.006)	(0.006)	(0.014)	(0.014)	(0.015)	(0.018)	(0.018)
Leverage	−0.024	−0.024	−0.022	−0.009	−0.009	−0.053	−0.053	−0.091	−0.158**	−0.158**
	(0.019)	(0.019)	(0.021)	(0.024)	(0.024)	(0.062)	(0.062)	(0.070)	(0.080)	(0.080)
Age	−0.010	−0.010	−0.006	−0.003	−0.003	0.018	0.018	0.010	0.015	0.015
	(0.007)	(0.007)	(0.007)	(0.009)	(0.009)	(0.017)	(0.017)	(0.021)	(0.028)	(0.028)
Profit	0.346***	0.346***	0.341***	0.335***	0.335***	−0.118	−0.118	−0.078	−0.028	−0.028
	(0.045)	(0.045)	(0.047)	(0.051)	(0.051)	(0.165)	(0.165)	(0.166)	(0.160)	(0.160)
Affected customer		0.004	0.003		0.002					
		(0.004)	(0.004)		(0.004)					
Affected supplier							−0.000	−0.003		−0.005
							(0.012)	(0.013)		(0.013)
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year-quarter FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
State x year FE	NO	NO	YES	Subsumed	Subsumed	NO	NO	YES	Subsumed	Subsumed
Industry x year FE	NO	NO	YES	Subsumed	Subsumed	NO	NO	YES	Subsumed	Subsumed
State x industry x year FE	NO	NO	NO	YES	YES	NO	NO	NO	YES	YES
Observations	107,106	107,106	93,702	93,450	93,450	107,106	107,106	93,702	93,450	93,450
R-squared	0.111	0.111	0.132	0.232	0.232	0.120	0.120	0.136	0.225	0.225

This table reports the effects of natural disasters on firms' receivables and payables. The unit of observation is a firm-quarter. The dependent variables are changes in receivables (columns 1 to 5) and payables (columns 6 to 10), defined as the difference between the ratio of receivables to sales and the ratio of payables to the cost of goods sold in quarter $t+4$ and t, respectively. The main independent variable is *Affected*, which is an indicator variable that equals one if a firm is located in a county that is impacted by a natural disaster in quarter t and zero otherwise. Firm controls include size, leverage, age, and profitability in quarter t. We also control for affected customer (supplier) when using change in receivables (payables) as the outcome variable. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

5.3 Tables 4–5: Pre-trends and bilateral supply-chain flows

Read:

- Table 4 finds no evidence that trade credit rose before disasters.
- Table 5 Panel A shows suppliers of affected firms increase receivables, and customers of affected firms increase payables.
- Table 5 Panel B uses bilateral SEC data and shows affected customers receive more trade credit and affected suppliers extend more trade credit.

Economic message: The bilateral evidence confirms the network mechanism: liquidity is extended by suppliers and reaches customers through the affected firm.

Table 4
Testing for pre-existing trends in firms' receivables and payables.

	Change in receivables [1]	Change in payables [2]
Disaster hits firm $_t$	0.018**	0.078***
	(0.007)	(0.024)
Disaster hits firm$_{t+5}$	−0.014	−0.034
	(0.009)	(0.023)
Disaster hits firm$_{t+6}$	−0.012	−0.031
	(0.010)	(0.033)
Disaster hits firm$_{t+7}$	−0.002	−0.013
	(0.010)	(0.027)
Disaster hits firm$_{t+8}$	−0.007	−0.006
	(0.008)	(0.025)
Disaster hits one customer (supplier) $_t$	0.003	−0.002
	(0.004)	(0.013)
Disaster hits one customer (supplier)$_{t+5}$	−0.000	−0.004
	(0.005)	(0.011)
Disaster hits one customer (supplier)$_{t+6}$	−0.003	0.004
	(0.005)	(0.012)
Disaster hits one customer (supplier)$_{t+7}$	−0.004	0.011
	(0.005)	(0.012)
Disaster hits one customer (supplier)$_{t+8}$	0.003	−0.006
	(0.004)	(0.012)
Controls	YES	YES
Firm FE	YES	YES
Year-quarter FE	YES	YES
State x year FE	YES	YES
Industry x year FE	YES	YES
Observations	93,702	93,702
R-squared	0.133	0.136

This table tests whether changes in firms' receivables and payables predate natural disasters. The unit of observation in each regression is a firm-quarter. The dependent variables are changes in receivables (column 1) and payables (column 2), defined as the difference between the ratio of receivables to sales and payables to the cost of goods sold in quarter $t + 4$ and t, respectively. The independent variables, *Disaster hits firm*$_t$, *Disaster hits firm*$_{t+5}$, *Disaster hits firm*$_{t+6}$, *Disaster hits firm*$_{t+7}$, and *Disaster hits firm*$_{t+8}$, are indicator variables that equal one if the firm is impacted by a natural disaster in the current, five, six, seven, and eight subsequent quarters, respectively, and zero otherwise. *Disaster hits one customer (supplier)*$_t$, *Disaster hits one customer (supplier)*$_{t+5}$, *Disaster hits one customer (supplier)*$_{t+6}$, *Disaster hits one customer (supplier)*$_{t+7}$, and *Disaster hits one customer (supplier)*$_{t+8}$, are indicator variables that equal one if one of their customers (suppliers) is impacted by a natural disaster in the current, five, six, seven, and eight subsequent quarters, respectively, and zero otherwise when using change in receivables (payables) as the outcome variable. Since disasters at five, six, seven, and eight subsequent quarters occur after the change in the dependent variable, i.e., between t and $t + 4$, their inclusion allows us to test for

(a) Original Table 4: Pre-trends.

Table 5
Natural disasters and affected firms' suppliers and customers.

Panel A: Compustat/Factset Revere sample

	Suppliers of affected firms Change in receivables [1]	Change in receivables [2]	Customers of affected firms Change in payables [3]	Change in payables [4]
Supplier of affected firm	0.013**	0.016**		
	(0.007)	(0.008)		
Customer of affected firm			0.036*	0.032*
			(0.019)	(0.019)
Affected		0.012*		0.077***
		(0.007)		(0.025)
Size	−0.014**	−0.014**	−0.007	−0.007
	(0.006)	(0.006)	(0.015)	(0.015)
Leverage	−0.027	−0.027	−0.117	−0.117
	(0.023)	(0.023)	(0.078)	(0.078)
Age	−0.003	−0.003	0.022	0.022
	(0.007)	(0.007)	(0.021)	(0.021)
Profit	0.706***	0.706***	−0.036	−0.035
	(0.075)	(0.075)	(0.225)	(0.225)
Firm FE	YES	YES	YES	YES
Year-quarter FE	YES	YES	YES	YES
State x year FE	YES	YES	YES	YES
Industry x year FE	YES	YES	YES	YES
Observations	85,372	85,372	86,346	86,346
R-squared	0.148	0.149	0.147	0.147

Panel B: SEC Customer-supplier-year level sample

Dependent variable	Identifying trade credit demand Change in trade credit [1]	Identifying trade credit supply Change in trade credit [2]
Affected customer	0.024***	
	(0.009)	
Affected supplier		0.021**
		(0.010)
Supplier x year FE	YES	
Customer x year FE		YES
Observations	853	1023
R-squared	0.623	0.340

This table reports estimates of the effects of natural disasters on the receivables of the suppliers and payables of the customers of affected firms. Panel A uses the merged Compustat/Factset Revere sample, whereas Panel B uses the SEC sample. In Panel A, the unit of observation in each regression is a firm-quarter. In Panel B, the unit of observation is a supplier-customer-year. The dependent variable in columns 1 and 2 of Panel A is the change in receivables, defined as the difference between the ratio of receivables to sales in quarter $t + 4$ and t, and in columns 3 and 4 of Panel A is the change in payables, defined as the difference between the ratio of payables to the cost of goods sold in quarter $t + 4$ and t. In Panel B, the dependent variable is the change in trade credit, defined as the difference between the amount of accounts receivable in year $t + 1$ and t granted by supplier s to firm i scaled by the sales amount of supplier s to firm i. In Panel A columns 1 and 2, the main independent variable is *Supplier of affected firm*, which is an indicator variable that equals one if the firm is a supplier of a firm that is impacted by a natural disaster at time t and zero otherwise, and in columns 3 and 4, is *Customer of affected firm*, which is an indicator variable that equals one if the firm is a customer of a firm that is impacted by a natural disaster at time t and zero otherwise. In Panel B columns 1 and 2, the main independent variables are *Affected customer*, which is an indicator variable that equals one if a firm's customer is located in a county that is impacted by a natural disaster in year t and zero otherwise, and *Affected supplier*, which is an indicator variable that equals one if a firm's supplier is located in a county that is impacted by a natural disaster in year t and zero otherwise, respectively. Firm controls include size, leverage, age, and profitability. All variables are defined in Appendix A. Robust standard errors clustered by firm in Panel A and by customer (supplier) firm in columns 1 (2) of Panel B are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

(b) Original Table 5: Suppliers and customers of affected firms.

5.4 Table 6: Customers' switching costs

Read:

- Product-market fluidity strengthens the trade-credit response of affected firms.
- Industry concentration weakens the response, consistent with higher switching costs in concentrated industries.
- In the SEC sample, large customers receive more trade credit from affected suppliers.

Economic message: Trade credit is larger when customers can credibly switch away from a disrupted supplier.

Table 6
Customers' switching costs.

Panel A: Compustat/Factset Revere sample	Change in receivables [1]	Change in payables [2]	Change in receivables [3]	Change in payables [4]
Affected	−0.017	−0.014	0.031**	0.126***
	(0.017)	(0.061)	(0.012)	(0.035)
Fluidity	0.002	−0.004		
	(0.001)	(0.004)		
Affected*Fluidity	0.004**	0.012*		
	(0.002)	(0.007)		
HHI			−0.002	0.097
			(0.029)	(0.062)
Affected*HHI			−0.086*	−0.328***
			(0.050)	(0.112)
Size	−0.021***	−0.006	−0.015***	−0.003
	(0.006)	(0.017)	(0.005)	(0.015)
Leverage	−0.011	−0.096	−0.022	−0.091
	(0.025)	(0.073)	(0.021)	(0.070)
Age	0.001	0.008	−0.005	0.010
	(0.008)	(0.025)	(0.007)	(0.021)
Profit	0.330***	−0.228	0.341***	−0.078
	(0.052)	(0.172)	(0.047)	(0.166)
Affected customer (supplier)	0.006	0.004	0.003	−0.003
	(0.004)	(0.015)	(0.004)	(0.013)
Firm FE	YES	YES	YES	YES
Year-quarter FE	YES	YES	YES	YES
State x year FE	YES	YES	YES	YES
Industry x year FE	YES	YES	YES	YES
Observations	78,176	78,176	93,702	93,702
R-squared	0.132	0.137	0.133	0.136

Panel B: SEC Customer-supplier-year level sample	Change in trade credit [1]
Affected supplier	−0.020
	(0.026)
Affected supplier*Customer size $_{25\_50}$	0.038
	(0.052)
Affected supplier*Customer size $_{50\_75}$	0.024
	(0.034)
Affected supplier*Customer size $_{75+}$	0.048*
	(0.027)
Customer x year FE	YES
Observations	1023
R-squared	0.341

This table reports estimates of the effects of natural disasters on the receivables and payables of firms whose customers face different switching costs. In Panel A, the unit of observation is a firm-quarter, using the merged Compustat/Factset Revere sample. The dependent variables are the changes in receivables (columns 1 and 3) and payables (columns 2 and 4), defined as the difference between the ratio of receivables to sales and the ratio of payables to the cost of goods sold in quarter $t+4$ and t, respectively. In columns 1 and 2, the measure of competitive environment is the product market fluidity proxy developed by Hoberg et al. (2014). In columns 3 and 4, the measure of competitive environment is the Herfindahl index (HHI) of the sales in the industry of a firm. The main independent variable is *Affected*, which is an indicator variable that equals one if a firm is located in a county that is impacted by a natural disaster in quarter t and zero otherwise. Firm controls include size, leverage, age, and profitability. We also control for affected customer (supplier) when using change in receivables (payables) as the outcome variable. In Panel B, the unit of observation is a supplier-customer-year, using the SEC sample. The dependent variable is the change in trade credit, defined as the difference between the amount of accounts

Original Table 6: Customers' switching costs.

5.5 Tables 7–8: Expected disruption and reputation

Read:

- Table 7 shows the trade-credit response is smaller for firms operating in more states and for firms with high inventories.

- Table 8 shows the response is smaller when the affected firm has longer customer relationships or superstar status.
- These characteristics make the supplier appear more reliable despite the shock.

Economic message: When customers expect less disruption, affected firms need to transfer less surplus through trade credit.

Table 7
Expected disruption.

	Change in receivables [1]	Change in payables [2]	Change in receivables [3]	Change in payables [4]
Affected	0.029**	0.113***	0.029**	0.126***
	(0.012)	(0.037)	(0.012)	(0.039)
Number of states	0.001	0.002*		
	(0.000)	(0.001)		
Affected*Number of states	−0.001**	−0.003**		
	(0.000)	(0.001)		
High inventory			0.001	−0.056***
			(0.007)	(0.020)
Affected*High inventory			−0.024*	−0.105***
			(0.014)	(0.039)
Size	−0.016***	−0.005	−0.015***	−0.006
	(0.005)	(0.015)	(0.005)	(0.015)
Leverage	−0.022	−0.092	−0.022	−0.091
	(0.021)	(0.070)	(0.021)	(0.070)
Age	−0.006	0.007	−0.006	0.011
	(0.007)	(0.021)	(0.007)	(0.021)
Profit	0.342***	−0.074	0.341***	−0.079
	(0.047)	(0.166)	(0.047)	(0.166)
Affected customer (supplier)	0.003	−0.003	0.003	−0.003
	(0.004)	(0.013)	(0.004)	(0.013)
Firm FE	YES	YES	YES	YES
Year-quarter FE	YES	YES	YES	YES
State x year FE	YES	YES	YES	YES
Industry x year FE	YES	YES	YES	YES
Observations	93,702	93,702	93,702	93,702
R-squared	0.133	0.136	0.133	0.136

This table reports estimates of the effects of natural disasters on the receivables and payables of firms with varying degrees of expected disruption. The unit of observation in each regression is a firm-quarter. The dependent variables are changes in receivables (columns 1 and 3) and payables (columns 2 and 4), defined as the difference between the ratio of receivables to sales and the ratio of payables to the cost of goods sold in quarter $t+4$ and t, respectively. *Number of states* indicates the number of states in which a firm operates. *High inventory* is an indicator variable that equals one if the firm has an above median level of inventories scaled by assets and zero otherwise. The main independent variable is *Affected*, which is an indicator variable that equals one if a firm is located in a county that is impacted by a natural disaster in quarter t and zero otherwise. Firm controls include size, leverage, age, and profitability. We also control for affected customer (supplier) when using change in receivables (payables) as the outcome variable. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

(a) Original Table 7: Expected disruption.

Table 8
Firm reputation.

	Change in receivables [1]	Change in payables [2]	Change in receivables [3]	Change in payables [4]
Affected	0.045**	0.152**	0.020***	0.086***
	(0.020)	(0.060)	(0.008)	(0.025)
Relationship length	−0.002*	−0.006		
	(0.001)	(0.004)		
Affected*Relationship length	−0.008*	−0.023*		
	(0.005)	(0.013)		
Superstar			0.046**	0.151**
			(0.018)	(0.074)
Affected*Superstar			−0.039**	−0.154***
			(0.018)	(0.052)
Size	−0.016***	−0.004	−0.016***	−0.004
	(0.005)	(0.015)	(0.005)	(0.015)
Leverage	−0.022	−0.092	−0.021	−0.089
	(0.021)	(0.070)	(0.021)	(0.070)
Age	−0.005	0.012	−0.006	0.009
	(0.007)	(0.021)	(0.007)	(0.021)
Profit	0.342***	−0.076	0.341***	−0.078
	(0.047)	(0.166)	(0.047)	(0.166)
Affected customer (supplier)	0.003	−0.003	0.003	−0.003
	(0.004)	(0.013)	(0.004)	(0.013)
Firm FE	YES	YES	YES	YES
Year-quarter FE	YES	YES	YES	YES
State x year FE	YES	YES	YES	YES
Industry x year FE	YES	YES	YES	YES
Observations	93,702	93,702	93,702	93,702
R-squared	0.133	0.136	0.133	0.136

This table reports estimates of the effects of natural disasters on the receivables and payables of firms with different reputation. The unit of observation in each regression is a firm-quarter. The dependent variables are changes in receivables (columns 1 and 3) and payables (columns 2 and 4), defined as the difference between the ratio of receivables to sales and the ratio of payables to the cost of goods sold in quarter $t+4$ and t, respectively. *Relationship length* is the average length of a treated firm's relationships with its customers, measured from when a firm and each of its customers are first reported in Factset Revere. *Superstar* is an indicator that equals one if the firm's market value is in the top 1% of the industry and zero otherwise. The main independent variable is *Affected*, which is an indicator variable that equals one if a firm is located in a county that is impacted by a natural disaster in quarter t and zero otherwise. Firm controls include size, leverage, age, and profitability. We also control for affected customer (supplier) when using change in receivables (payables) as the outcome variable. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

(b) Original Table 8: Firm reputation.

5.6 Tables 9–10: Stakes in the production network

Read:

- Table 9 shows affected firms extend more trade credit when they depend more on customers, especially major customers.
- Table 10 shows suppliers of affected firms extend more trade credit when their own stakes in the affected firm and its customers are larger.

- The mechanism works through both direct and indirect dependence.

Economic message: The more valuable the supply-chain relationship is, the stronger the incentive to use trade credit to keep the chain intact.

Table 9
Natural disasters and affected firms' dependence on customers.

	Change in receivables [1]	Change in payables [2]	Change in receivables [3]	Change in payables [4]
Affected	0.009	0.051**	0.003	−0.004
	(0.006)	(0.020)	(0.010)	(0.006)
Relative size	0.001	−0.002		
	(0.001)	(0.004)		
Affected*Relative size	0.007*	0.019*		
	(0.004)	(0.011)		
Major customer			0.012*	0.004
			(0.006)	(0.004)
Affected*Major customer			0.054**	0.022**
			(0.021)	(0.011)
Size	−0.015***	−0.005	−0.016***	−0.009***
	(0.005)	(0.015)	(0.005)	(0.003)
Leverage	−0.022	−0.091	−0.022	−0.013
	(0.021)	(0.070)	(0.021)	(0.010)
Age	−0.006	0.009	−0.005	0.004
	(0.007)	(0.021)	(0.007)	(0.004)
Profit	0.341***	−0.076	0.340***	−0.010
	(0.048)	(0.166)	(0.047)	(0.025)
Affected customer (supplier)	0.004	−0.001	0.003	0.001
	(0.004)	(0.012)	(0.004)	(0.003)
Firm FE	YES	YES	YES	YES
Year-quarter FE	YES	YES	YES	YES
State x year FE	YES	YES	YES	YES
Industry x year FE	YES	YES	YES	YES
Observations	93,702	93,702	93,702	93,702
R-squared	0.133	0.136	0.133	0.122

This table reports estimates of the effects of natural disasters on the receivables and payables of affected firms that depend on their customers to different extents. The unit of observation in each regression is a firm-quarter. The dependent variables are the changes in receivables (columns 1 and 3) and payables (columns 2 and 4), defined as the difference between the ratio of receivables to sales and the ratio of payables to the cost of goods sold in quarter $t + 4$ and t, respectively. In columns 1 and 2, the measure of dependence is *Relative size*, calculated as the natural logarithm of the affected firm's customers' average size scaled by its own size. In columns 3 and 4, the measure of dependence is *Major customer*, which is an indicator variable that equals one if the firm reports a major customer in its financial statements and zero otherwise. The main independent variable is *Affected*, which is an indicator variable that equals one if a firm is located in a county that is impacted by a natural disaster in quarter t and zero otherwise. Firm controls include size, leverage, age, and profitability. We also control for affected customer (supplier) when using change in receivables (payables) as the outcome variable. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

(a) Original Table 9: Dependence on customers.

Table 10
Natural disasters and trade credit along the supply chain.

	Change in receivables [1]	Change in receivables [2]
Supplier of affected firm	0.012*	0.001
	(0.007)	(0.007)
Relative size $_{1\_2}$	−0.002	
	(0.002)	
Relative size $_{0\_1}$	−0.001	
	(0.001)	
Supplier of affected firm*Relative size $_{1\_2}$	0.007*	
	(0.004)	
Supplier of affected firm*Relative size $_{0\_1}$	0.007**	
	(0.004)	
Major customer $_{1\_2}$		0.014**
		(0.006)
Major customer $_{0\_1}$		0.001
		(0.013)
Supplier of affected firm*Major customer $_{1\_2}$		0.025*
		(0.015)
Supplier of affected firm*Major customer $_{0\_1}$		0.081**
		(0.040)
Firm controls	YES	YES
Firm FE	YES	YES
Year-quarter FE	YES	YES
State x year FE	YES	YES
Industry x year FE	YES	YES
Observations	85,372	85,372
R-squared	0.149	0.149

This table reports estimates of the effects of natural disasters on the receivables of the suppliers of affected firms. The unit of observation in each regression is a firm-quarter. The dependent variable is the change in receivables, defined as the difference between the ratio of receivables to sales. In all columns, the main independent variable is *Supplier of affected firm*, which is an indicator variable that equals one if the firm is a supplier of a firm that is impacted by a natural disaster at time t and zero otherwise. In column 1, the cross-sectional variable is *Relative size*, calculated as the natural logarithm of the average size of firm's customers scaled by the firm's size. *Relative size* $_{1\_2}$ considers the relative size of the affected firms and their suppliers. Similarly, *Relative size* $_{0\_1}$ considers the relative size of the affected firms and their customers. In column 2, the cross-sectional variable is *Major customer. Major customer* $_{1\_2}$ is an indicator variable that equals one if the supplier of an affected firm reports the affected firm as a major customer and zero otherwise. Similarly, *Major customer* $_{0\_1}$ is an indicator variable that equals one if the affected firm reports one of its customers as a major customer. Firm controls include size, leverage, age, profitability, and whether the firm is affected. All variables are defined in Appendix A. Robust

(b) Original Table 10: Trade credit along the supply chain.

5.7 Tables 11–13: Financial constraints and supply-chain instability

Read:

- Table 11 shows affected firms' ability to expand trade credit is limited when their suppliers are financially constrained.
- Table 12 shows customers are more likely to sever relationships or find new suppliers when financial constraints are pervasive.
- Table 13 shows sales and profitability decline when affected firms and their suppliers are financially constrained.

Economic message: Trade credit stabilizes supply chains only when financially unconstrained firms upstream can provide liquidity. When constraints bind throughout the chain, customer-supplier links break and performance falls.

Table 11
Financial constraints and trade credit.

Panel A: Financial constraint proxy - market capitalization below the median

Subsample	Constrained firms		Unconstrained firms	
Dependent variable	Change in receivables [1]	Change in payables [2]	Change in receivables [3]	Change in payables [4]
Affected	0.054*	0.281**	0.018**	0.056***
	(0.033)	(0.116)	(0.008)	(0.021)
Percentage of constrained suppliers	0.037	0.121	0.003	−0.022
	(0.051)	(0.140)	(0.009)	(0.022)
Affected* Percentage of constrained suppliers	−0.249*	−0.522**	−0.035	−0.101
	(0.141)	(0.222)	(0.030)	(0.068)
Size	−0.053***	0.028	−0.019***	−0.023
	(0.020)	(0.067)	(0.005)	(0.016)
Leverage	0.045	−0.344*	−0.025	−0.069
	(0.053)	(0.199)	(0.022)	(0.069)
Age	−0.046	−0.117	0.003	0.038**
	(0.036)	(0.135)	(0.006)	(0.017)
Profit	0.307***	0.239	0.371***	−0.203
	(0.076)	(0.243)	(0.066)	(0.191)
Affected customer (supplier)	0.014	−0.053	0.001	0.015
	(0.020)	(0.046)	(0.004)	(0.011)
Firm FE	YES	YES	YES	YES
Year-quarter FE	YES	YES	YES	YES
State x year FE	YES	YES	YES	YES
Industry x year FE	YES	YES	YES	YES
Observations	18,141	18,141	74,748	74,748
R-squared	0.235	0.270	0.165	0.145

Panel B: Financial constraint proxy - no bond rating

Subsample	Constrained firms		Unconstrained firms	
Dependent variable	Change in receivables [1]	Change in payables [2]	Change in receivables [3]	Change in payables [4]
Affected	0.066***	0.192**	0.036*	0.075*
	(0.025)	(0.078)	(0.021)	(0.043)
Percentage of constrained suppliers	0.023*	−0.036	0.003	−0.025
	(0.013)	(0.036)	(0.009)	(0.023)
Affected* Percentage of constrained suppliers	−0.090*	−0.235**	−0.066	−0.038
	(0.052)	(0.116)	(0.042)	(0.094)
Size	−0.018**	0.005	−0.020***	−0.087***
	(0.007)	(0.019)	(0.006)	(0.024)
Leverage	−0.023	−0.128	−0.026	−0.007
	(0.030)	(0.096)	(0.021)	(0.071)
Age	−0.014	−0.007	0.013	0.046*
	(0.009)	(0.030)	(0.011)	(0.027)
Profit	0.349***	0.168	0.278***	−1.483***
	(0.055)	(0.185)	(0.072)	(0.286)
Affected customer (supplier)	0.000	−0.001	0.005	0.015
	(0.007)	(0.019)	(0.004)	(0.012)
Firm FE	YES	YES	YES	YES
Year-quarter FE	YES	YES	YES	YES
State x year FE	YES	YES	YES	YES
Industry x year FE	YES	YES	YES	YES
Observations	59,200	59,200	34,096	34,096
R-squared	0.151	0.164	0.173	0.172

This table reports estimates of the effects of natural disasters on firms' receivables and payables using different measures of financial constraints for firms and their suppliers. In Panel A, we classify a firm and its suppliers as financially unconstrained if their size (measured by market capitalization) is above the median. In Panel B, we classify a firm and its suppliers as financially unconstrained if it has a bond rating. Since firms report several suppliers, we calculate the percentage of suppliers that are financially constrained. The unit of observation in each regression is a firm-quarter. The dependent variables are the

Original Table 11: Financial constraints and trade credit.

5.8 Tables 14–15: Extensions

Read:

- Table 14 shows firms exposed to operating risk and dependent on customers tend to choose larger suppliers.

Table 12
Financial constraints and changes in customer-supplier relationships.

Panel A: Dependent variable - Change in number of customers for the affected firms

Financial constraint proxy	Market capitalization below the median		No bond rating	
Subsample	Constrained Firms	Unconstrained Firms	Constrained Firms	Unconstrained Firms
	[1]	[2]	[3]	[4]
Affected	0.013	0.001	0.014	0.002
	(0.013)	(0.006)	(0.014)	(0.016)
Percentage of constrained suppliers	−0.033	−0.005	0.012	−0.004
	(0.040)	(0.016)	(0.016)	(0.017)
Affected*	−0.205*	−0.030	−0.049*	−0.002
Percentage of constrained suppliers	(0.106)	(0.037)	(0.029)	(0.032)
Size	0.016	0.015**	0.004	0.046***
	(0.015)	(0.007)	(0.007)	(0.012)
Leverage	0.042	0.003	0.005	−0.010
	(0.038)	(0.023)	(0.023)	(0.036)
Age	−0.040	−0.066***	−0.062***	−0.063***
	(0.035)	(0.012)	(0.013)	(0.022)
Profit	0.001	0.106**	−0.010	0.226***
	(0.039)	(0.044)	(0.033)	(0.066)
Firm FE	YES	YES	YES	YES
Year-quarter FE	YES	YES	YES	YES
State x year FE	YES	YES	YES	YES
Industry x year FE	YES	YES	YES	YES
Observations	14,621	67,355	51,499	30,815
R-squared	0.503	0.276	0.303	0.376

Panel B: Dependent variable - Change in number of new suppliers for the affected firms' customers

Financial constraint proxy	Market capitalization below the median		No bond rating	
Subsample	Constrained Firms	Unconstrained Firms	Constrained Firms	Unconstrained Firms
	[1]	[2]	[3]	[4]
Customer of affected firm	−0.025	0.006	−0.032	0.035
	(0.016)	(0.011)	(0.022)	(0.027)
Percentage of constrained suppliers	−0.012	−0.024	0.021	−0.027
	(0.039)	(0.030)	(0.020)	(0.025)
Customer of affected firm*	0.097**	0.091	0.077*	−0.025
Percentage of constrained suppliers	(0.049)	(0.071)	(0.044)	(0.047)
Size	−0.002	−0.009	−0.005	−0.007
	(0.014)	(0.006)	(0.007)	(0.006)
Leverage	0.005	−0.034	−0.038	−0.042*
	(0.058)	(0.024)	(0.029)	(0.024)
Age	−0.019	−0.001	−0.014	0.012
	(0.032)	(0.011)	(0.013)	(0.012)
Profit	−0.252	0.061	0.059	0.039
	(0.158)	(0.063)	(0.076)	(0.066)
Firm FE	YES	YES	YES	YES
Year-quarter FE	YES	YES	YES	YES
State x year FE	YES	YES	YES	YES
Industry x year FE	YES	YES	YES	YES
Observations	19,514	84,601	61,551	76,178
R-squared	0.400	0.172	0.213	0.19

This table reports estimates of the effects of natural disasters on firms' number of customers (Panel A) and customers' number of new suppliers (Panel B). In columns 1 and 2, we classify a firm as financially unconstrained if its size (measured by market capitalization) is above the median. In columns 3 and 4, we classify a firm as financially unconstrained if it has a bond rating. Since firms report several suppliers, we calculate the percentage of suppliers that are financially constrained. The unit of observation in each regression is a firm-quarter. In columns 1 and 3, we consider the subsample of financially constrained firms. In columns 2 and 4, we consider the subsample of financially unconstrained firms. The main independent variables in Panels A and B are *Affected*, which is an indicator variable that equals one if a firm is located in a county that is impacted by a natural disaster in quarter t and zero otherwise, and *Customer of affected firm*, which is an indicator variable that equals one if the firm is a customer of a firm that is impacted by a natural disaster at time t and zero otherwise, respectively. Firm controls include size, leverage, age, and profitability. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

Table 13
Natural disasters, financial constraints, and affected firms' performance.

Panel A: Dependent variable - Change in ROA

Financial constraint proxy	Market capitalization below the median		No bond rating	
Subsample	Constrained Firms	Unconstrained Firms	Constrained Firms	Unconstrained Firms
	[1]	[2]	[3]	[4]
Affected	0.001	0.001	0.003	0.000
	(0.003)	(0.001)	(0.002)	(0.002)
Percentage of constrained suppliers	0.002	0.001	−0.002*	−0.001*
	(0.003)	(0.001)	(0.001)	(0.001)
Affected*	−0.019***	0.002	−0.005*	0.001
Percentage of constrained suppliers	(0.007)	(0.003)	(0.003)	(0.003)
Size	−0.004***	−0.003***	−0.003***	−0.004***
	(0.001)	(0.000)	(0.001)	(0.001)
Leverage	0.004	0.005***	0.005**	0.008***
	(0.005)	(0.002)	(0.002)	(0.002)
Age	−0.001	0.001	0.001	0.000
	(0.003)	(0.001)	(0.001)	(0.001)
Profit	−0.146***	−0.199***	−0.166***	−0.180***
	(0.009)	(0.008)	(0.007)	(0.013)
Firm FE	YES	YES	YES	YES
Year-quarter FE	YES	YES	YES	YES
State x year FE	YES	YES	YES	YES
Industry x year FE	YES	YES	YES	YES
Observations	18,141	74,748	59,200	34,096
R-squared	0.311	0.271	0.235	0.329

Panel B: Dependent variable - Change in log sales

Financial constraint proxy	Market capitalization below the median		No bond rating	
Subsample	Constrained Firms	Unconstrained Firms	Constrained Firms	Unconstrained Firms
	[1]	[2]	[3]	[4]
Affected	0.022	0.001	0.032***	−0.016
	(0.014)	(0.005)	(0.010)	(0.010)
Percentage of constrained suppliers	−0.003	0.010	−0.016*	−0.003
	(0.023)	(0.007)	(0.009)	(0.009)
Affected*	−0.093**	0.017	−0.043**	0.021
Percentage of constrained suppliers	(0.045)	(0.026)	(0.020)	(0.019)
Size	−0.076***	−0.080***	−0.065***	−0.088***
	(0.009)	(0.005)	(0.005)	(0.008)
Leverage	−0.055**	−0.032**	−0.039**	−0.030
	(0.022)	(0.014)	(0.015)	(0.020)
Age	−0.046**	−0.015**	−0.032***	−0.023**
	(0.020)	(0.006)	(0.008)	(0.011)
Profit	−0.265***	−0.401***	−0.323***	−0.169***
	(0.038)	(0.045)	(0.033)	(0.050)
Firm FE	YES	YES	YES	YES
Year-quarter FE	YES	YES	YES	YES
State x year FE	YES	YES	YES	YES
Industry x year FE	YES	YES	YES	YES
Observations	18,141	74,748	59,200	34,096
R-squared	0.415	0.380	0.345	0.461

This table reports estimates of the effects of natural disasters on firms' ROA (Panel A) and Sales (Panel B) using different measures of financial constraints for firms and their suppliers. In columns 1 and 2, we classify a firm as financially unconstrained if its size (measured by market capitalization) is above the median. In columns 3 and 4, we classify a firm as financially unconstrained if it has a bond rating. Since firms report several suppliers, we calculate the percentage of suppliers that are financially constrained. The unit of observation in each regression is a firm-quarter. The dependent variables are the change in ROA and the natural logarithm of sales, defined as the difference between the value in quarter $t + 4$ and t. In columns 1 and 3, we consider the subsample of financially constrained firms. In columns 2 and 4, we consider

- Table 15 uses textual supply-chain risk from earnings calls and finds that high-risk firms increase both receivables and payables.
- This supports external validity beyond natural disasters.

Economic message: Firms appear to manage operating risk by choosing stronger suppliers and by using trade credit when supply-chain risk rises.

Table 14
Natural disasters and supplier choice.

Measure of dependence on customer	Relative size	Major customer
	Avg supplier size by industry	Avg supplier size by industry
	[1]	[2]
Size	−0.026	−0.017
	(0.021)	(0.016)
Disaster in last 5 years	−0.341***	−0.082
	(0.107)	(0.070)
Size*Disaster in last 5 years	0.039***	0.014
	(0.013)	(0.010)
Customer dependence	−0.028*	−0.101*
	(0.016)	(0.056)
Customer dependence*Size	0.006**	0.022**
	(0.002)	(0.009)
Customer dependence*Disaster in last 5 years	0.067***	0.171*
	(0.021)	(0.098)
Size*Customer dependence* Disaster in last 5 years	−0.007**	−0.028*
	(0.003)	(0.014)
Leverage	0.057	0.053
	(0.070)	(0.056)
Profit	−0.014	−0.032
	(0.044)	(0.036)
Age	−0.141***	−0.115***
	(0.030)	(0.023)
Firm FE	YES	YES
Industry x year FE	YES	YES
Supplier Industry x year FE	YES	YES
Observations	180,815	235,989
R-squared	0.606	0.613

This table reports estimates of the effects of natural disasters on firms' supplier choice. The unit of observation is a firm-supplier industry-year. The dependent variable *Avg supplier size by industry* is the average size of the suppliers in a given year and industry for the firm. *Size* is the firm's natural logarithm of total assets. *Disaster in last 5 years* is an indicator variable that equals one if the firm is in a county that had a natural disaster in the last 5 years and zero otherwise. We measure a firm's *Customer dependence* using two measures: relative size (column 1) and major customer (column 2). *Relative size* is the natural logarithm of the affected firm's customers' average size scaled by its own size; *Major customer* is an indicator variable that equals one if the firm reports a major customer in its financial statements and zero otherwise. Firm controls include leverage, age, and profitability. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

7.2. Other operating shocks and comparison with liquidity shocks

We expect trade credit flows to increase when operating difficulties
for a firm in the network imperil the survival of the supply chain. Not

(a) Original Table 14: Natural disasters and supplier choice.

Table 15
The impact of supply chain risk on firms' receivables and payables.

	Change in receivables	Change in payables
	[1]	[2]
High SCRisk	0.006*	0.024**
	(0.003)	(0.012)
Size	−0.018***	−0.045**
	(0.006)	(0.019)
Leverage	0.001	−0.101
	(0.027)	(0.082)
Age	0.004	0.023
	(0.007)	(0.021)
Profit	0.373***	0.002
	(0.063)	(0.210)
Firm FE	YES	YES
Year-Quarter FE	YES	YES
State x year FE	YES	YES
Industry x year FE	YES	YES
Observations	56,367	56,367
R-squared	0.165	0.149

This table reports the effects of supply chain risk on firms' receivables and payables. The unit of observation is a firm-quarter. The dependent variables are changes in receivables (column 1) and payables (column 2), defined as the difference between the ratio of receivables to sales and the ratio of payables to the cost of goods sold in quarter $t + 4$ and t, respectively. The main independent variable is *High SCRisk*, which is an indicator variable that equals one if the firm's supply chain risk is in the top quintile and zero otherwise, according to the supply chain risk measure in Ersahin et al. (2024). Firm controls include size, leverage, age, and profitability in quarter *t*. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

after liquidity shortfalls that do not impair its ability to produce. In this

(b) Original Table 15: Supply-chain risk and trade credit.

6 Short summary

- The paper studies whether trade credit stabilizes production networks after operating shocks.
- Natural disasters are used as plausibly exogenous shocks to supplier reliability.
- Affected firms receive more trade credit from suppliers and provide more trade credit to customers.
- The response is stronger when customers can more easily switch suppliers.
- The response is weaker when disruption is expected to be smaller because the firm has more locations, inventories, reputation, or long relationships.
- Suppliers of affected firms also extend more trade credit when they have more to lose from a broken downstream relationship.
- Financial constraints upstream reduce trade-credit flows.
- When both affected firms and their suppliers are constrained, customers sever links and affected firms' sales and profitability fall.

- The evidence supports the view that trade credit is a contractual tool for preserving supply chains, not just a financing instrument.

7 Conclusion

7.1 Core conclusion

Ersahin, Giannetti, and Huang show that trade credit helps preserve customer-supplier relationships after operating shocks. Natural disasters make affected firms less reliable suppliers. To prevent customers from switching away, affected firms extend more trade credit, and their own suppliers provide more trade credit to make that extension feasible.

7.2 Economic intuition

Trade credit reallocates surplus inside the production network. When a customer may abandon a disrupted supplier, delayed payment terms raise the value of staying. Upstream suppliers support this transfer because they also benefit from keeping the downstream relationship alive. This turns trade credit into a supply-chain stabilization mechanism.

7.3 Takeaway

The paper broadens the interpretation of trade credit. It is not merely a substitute for bank finance or a bilateral bargaining outcome. It is a network contract that can keep production chains intact after operating shocks. The policy implication is that preserving liquidity in supply chains can support real production, especially when operating disruptions threaten many connected firms.